Dynamic Crime Hotspot and Crime Volume Prediction Using Spatio-Temporal Features

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ABSTRACT

Crime prediction is a very important way for crime prevention to reduce the losses of public property and life. Dynamic crime hotspot prediction and future crime count estimation (crime volume prediction) are two very popular analytical technique for crime prediction. Policing and other crime reduction resources can be deployed more efficiently based on these prediction results. In this paper, we propose a novel machine-learning based two-stage solution for dynamic crime hotspot prediction and crime volume prediction. We developed our solution and tested it on Denver crime dataset and found that our approach improves performance by 5% over baseline binary hotspot prediction.

KEYWORDS

Crime Prediction, Crime Hotspot Prediction, Crime Volume Prediction, Spatio-temporal Event

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1 INTRODUCTION

Crime in any form is a major social problem for public safety. A key component for tackling crime problems involves the analysis of where and when crimes taking place. Crime also does not occur randomly. It tends to concentrate at particular places due to the factors associated to the victim and the offender interaction as well as the opportunities that exist to commit crime. These concentrations or clusters of crime are commonly referred to as crime hotspots. Hotspot mapping has become a popular analytical technique used by the law enforcement, police and crime reduction agencies to visually identify where crime tends to be the highest. This helps them in decision-making that determines where to target and deploy resources within a given time duration [3] [2] [8]. Given limited policing resources, it is important to find optimal deployment strategies to minimize the response time as well as to maximize the coverage for police assistance. On the other hand, crime volume prediction is an estimation of the future crime count for a geographical area. Using the crime volume prediction over different parts of an a geographical region, prediction of crime hotspots can be obtained easily. Apart from police patrolling design, crime volume and hotspot prediction results can be useful for other usecases, such as, for a tourist portal, where a traveller can get to know about the crime risk profile of different areas in a region over time helping him a better future trip planning.

Existing works on crime prediction use historical crime records [2] [3] [8] as well as other data sources related to relevant socio-economic factors. The researchers found that crime in the real-world highly correlates with time, place and population. They have analyzed the correlations between the crime events and other socio-economic factors, such as, demographics [2], education [7], unemployment [19], ethnicity [18], urban infrastructures [10] etc. They leveraged these factors to improve crime hotspot prediction accuracy on top of the historical crime record data. But there are some challenges and limitations of using these additional information in crime prediction

- The data sources for these socio-economic factors are not updated very frequently and periodically. For example, demographic data in the United States are mostly collected and reported on yearly basis[1].
- These socio-economic data are mostly collected at county/block/school-district level which is much bigger than the jurisdiction area of a police precinct. These coarser level of data is not that much useful.
- The use of demographic, ethnicity, unemployment related data in crime prediction may lead to the privacy or civil rights violations [17].

In this paper, we propose a machine learning based two stage solution for crime volume prediction and crime hotspot prediction. We use only historical crime data to generate the predictions. We first divide the geographical area into square size grids. With this settings, except the grids at the region boundary, all the grids created will be of the same size. Our proposed model generates crime volume and crime hotspot prediction at the grid level for a given future time window. Here we have implemented a monthly prediction model due to sparsity of the crime incident data. If we go beyond month, e.g., week or day, then we don't get sufficient data points at the grid level to build a generalized and robust model. Some of the existing deep-learning based approach [13] [22] has

[1] https://www.census.gov/ces/dataproducts/demographicdata.html

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reported high accuracy numbers in crime hotspot prediction. But we didn't pursue deep-learning for our crime prediction task due to the following reasons

- Sparsity of crime data for the grids at monthly level.
- Performance challenge in a enterprise system setting for generalized data.

In the first stage of our solution we predict the crime volume at the grid level for future 1 month. Here instead of giving an absolute crime count we predict a volume bucket label corresponding a range of crime count. These volume buckets are pre-defined by us according to the range of crime count. In this way, we model the crime volume prediction problem as multi-class classification problem. To the best our knowledge, this has not been considered before in the context of crime prediction. In the second stage of our solution, we have used the predicted crime volume to predict the hotspot status of the grids in the future time window. We have developed our solution and tested it on a real life dataset (Denver crime data). We compared our results with a baseline binary hotspot prediction approach [2]. We take it as our baseline, partly for its performance and partly due to the absence of any standard benchmark. We find that our approach improves the performance by 5% over the baseline.

The rest of the article is organized as follows: Section 2 presents related work. Sections 3 illustrates our approach for crime volume and hotspot prediction. Section 4 describes the dataset we have used for this study. Section 5 describes the subsequent data pre-processing done. Sections 6 discusses about the experiment we performed and the results obtained. We conclude our work in Section 7.

2 RELATED WORK

The existing crime prediction methods have applied a variety of machine learning techniques, such as, time series analysis [9], regression analysis [6] [4], kernel density estimation(KDE) [1] [8], support vector machine (SVM)[11] and self-exciting point processes [14] [20] [16] [15] on historical crime data, socio-economic data, social media data etc.

Liao et al. [12] built a Bayesian-based crime prediction model using geographical information and victim characteristics. They divided characteristics of crime sites into two regional types: private regions and public regions. Next, they used a discrete distance decay function to create a geographic profile, which is the probability distribution of crime occurrences. Finally, to accurately predict the location of the next crime occurrence, geographic profiles were combined with Bayesian learning theory. Gorr et al. [9] proposed a short-term crime prediction method using a one-month time horizon. They used statistical crime data as well as demographic, social, and education information for crime prediction. Similarly, Chen et al.[4] applied an auto-regressive integrated moving average model for short-term crime prediction in a city in China. Shingleton et al. [21] used an approach based on regression analysis to predict three crime types (violence, homicide, and assault) in Salinas, California using ordinary least squares, Poisson regression, and negative binomial regression models. For prediction of crime hotspots, Kianmehr and Alhajj [11] proposed a computational framework for application in Columbus, Ohio and St. Louis, Missouri using SVM with k-means clustering. Furthermore, Wang et al. [23] used SVM for the prediction of criminal recidivism. They used datasets from the National Archive of Criminal Justice Data of the Inter-University Consortium for Political and Social Research. Recently, some studies have used social media data and KDE for crime prediction [8] [5]. Gerber [8], for example, predicted crimes using social media information (e.g., GPS-tagged tweets).

Despite all these contributions, most of the existing methods treat data from multiple domains on an equal footing, such as by directly concatenating features or performing a weighted summation. These approaches do not consider the different characteristics of data from multiple domains, which can be problematic. Also considering socio-economic factor(e.g., demographic, ethnicity, unemployment etc.) in crime prediction may lead to the privacy or civil rights violation issues [17].

3 OUR APPROACH

In our proposed method we mainly focus on the spatial and temporal aspects of crime data. Our two stage hierarchical approach predicts crime volume and crime hotspots using only spatio-temporal features derived from historical crime data. The system architecture of our proposed crime prediction solution is shown in Figure 1. It is divided into two modules: 1) GIS module 2) Machine Learning (ML) module. The approach starts at GIS module with the Data Management block creating a grid layer on top of the geographical area of our study and assigns the historical crime event data to the generated grids. Spatial and temporal aggregation of the crime data are done in this step. The system is designed in such a way that the grid size is made configurable and we carried out the experiments for different grid configurations. In the Model Building block, first the spatial and temporal features are calculated from aggregated data and then the featurized crime data is used to build crime volume prediction model using off-the-shelf ML algorithm. The crime volume prediction labels are given as the input to the hotspot prediction module and it predicts the top N(number of hotspots to be predicted) hotspots for the next month. N is given as user input. For better visualization of the predicted crime volume and crime hotspots, the results are sent back to the GIS component. The visualization improves the user understanding on future crime state of the region being analyzed. The steps followed in our crime volume and crime hotspot prediction is shown in the workflow table in the next page.

3.1 Crime Volume Prediction

In crime volume prediction, instead of predicting the grid level absolute crime count, we predict the crime volume bucket

[2] https://www.prnewswire.com/news-releases/conduents-crime-forecasting-analytics-wins-us-department-of-justice-challenge-prize-300650164.html

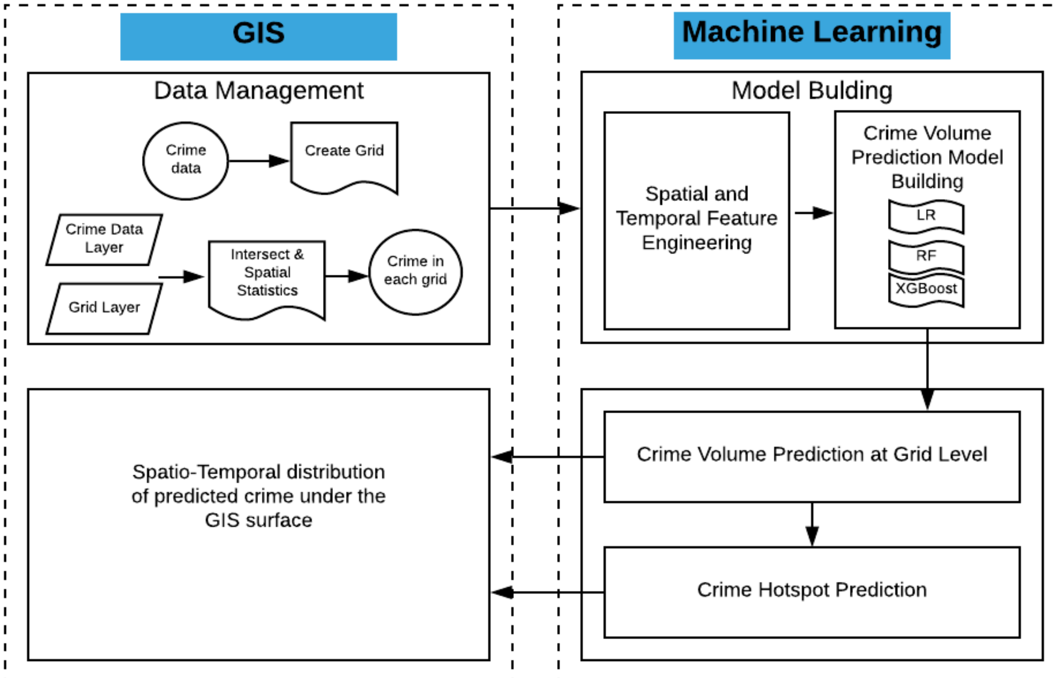

Figure 1: Crime Prediction System Architecture

Workflow: Crime Volume and Crime Hotspot Prediction

- Create square grids in our region of study. Map the crime data points to the grids using latitude, longitude information.
- Aggregate the crime spatio-temporally(grid and month wise). Get grid level crime count on monthly basis. Divide the crime count with grid area to calculate monthly normalized crime count.
- Create spatio-temporal features for the crime data.
- Build machine learning model for crime volume bucket prediction for grids.
- Predict crime hotspots using the grid level crime volume prediction.

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to which the grid will belong in next month. In this way, we solve the crime volume prediction problem as multi-class classification problem. First we discretized the range of grid level crime count into a finite number of volume buckets. The number of volume buckets created depends on historical crime count distribution. Then we have followed a two stage hierarchical classification approach to build crime volume prediction models using set of spatio-temporal features engineered from the historical crime event data. The formulation of the problem as a multi-class classification task is driven

Algorithm 1: Two-stage Crime Volume Prediction

Input: *Grids with spatio-temporal features*
Output: *Grids with predicted crime volume bucket*
Build Stage1(zero/non-zero) volume prediction model.
Build Stage2(1,..., K) volume prediction model with training dataset having non-zero Stage1 bucket label.
For test dataset first get the first level prediction using Stage1 classifier. Remove the zero prediction. Predict second level volume prediction using Stage2 classifier.

by several motivations. First, an important advantage of discretizing the volume prediction is that it greatly simplifies the presentation of the results so that it can be easily understood by wide number of audiences. Secondly, we have observed that the historical crime count follows a highly skewed distribution and categorizing a continuous variable is statistically convenient when dealing with such highly skewed distributions. Our two-stage crime volume prediction approach is described in Algorithm 1.

3.2 Crime Hotspot Prediction

In crime hotspot prediction we have used the predicted crime volume bucket. The number of hotspots(N) to be predicted is taken as user input. We predict the hotspot label for the grids using the following Algorithm 2.

Algorithm 2: Crime Hotspot Prediction Using Predicted Crime Volume Label

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Input: Grid level volume prediction for next month, Hotspot count(N)
Output: Grids predicted as hotspot for the next month
For all crime data points set class probability to max of individual bucket probabilities.
crime bucket K: data points with predicted volume = K.
crime bucket combined: stack up the crime buckets in the bucket order K, K-1, ..., 1
For first N data points in crime bucket combined set Hotspot Predicted = 1
```

Table 1: Category-wise Crime Distribution

Crime Category	Count	Percentage
aggravated-assault	10659	3.08%
all-other-crimes	82326	23.81%
arson	608	0.18%
auto-theft	23942	6.92%
burglary	24877	7.19%
drug-alcohol	31259	9.04%
larceny	47949	13.87%
murder	277	0.08%
other-crimes-against-persons	22480	6.50%
public-disorder	50756	14.68%
robbery	6158	1.78%
sexual-assault	3667	1.06%
theft-from-motor-vehicle	34680	10.03%
white-collar-crime	6150	1.78%

4 DATASET

In this work, we have used historical crime event data as the only source of information for building the prediction model. We have taken crime data for Denver for our experimentation purpose. This dataset includes all reported crimes in Denver city and county, Colorado for 5 years (2013-2018). The dataset information is based on the National Incident Based Reporting System (NIBRS)[3]. This dataset contains crime timestamp, crime location (longitude, latitude, and address), neighborhood, precinct ID, crime category (e.g., arson, auto-theft, burglary, murder, sexual-assault etc.). A subset of the Denver crime dataset is traffic accident instances. In our experiment we have excluded the traffic accident incidences from the dataset. Table 1 gives the related statistics of the Denver crime data in different crime categories.

5 PREPROCESSING

Once we got the data we carried out a series of pre-processing steps on the raw data. Tough we have given category-wise crime distribution in Table 1, we have considered all the different crime categories as same in our experiment due of the sparseness of the categorical crimes. In our experiment we have done the crime volume and crime hotspot prediction on monthly basis. First we created grids in the geographical area of our study using the GIS tool QGIS. Here one of the challenge is to keep the grid size small and still get a reasonably good estimate for crime volume and hotspot. We experimented with different grid size and still 1 sq-km is giving reasonable results. Taking grid size as 1 sq-km is also reasonable since the jurisdiction area of a police precinct is more likely in that range. Then we map the point crime data to grids using latitude longitude information. Each crime point is thus mapped to some grid id. Then we aggregated the crime data spatio-temporally(grid and month) and calculated the monthly crime count per grid. We divided it by grid area to get normalized crime count per grid on monthly basis. We call it as crime volume of that grid for that month and it serves as a proxy for crime intensity of that grid over the temporal period. Figure 2 shows grid level crime volume distribution which is very much skewed in nature. It says that majority of grids in our dataset have low crime volume (e.g., only 1 crime event per month), while in a small proportion of the grids crime volume is high. Given the high skewness of the distribution and to solve the crime volume prediction problem as a multi-class classification problem we discritized the range of crime volume value into a finite number of buckets.

For Denver dataset, we have created 6 buckets for crime volume prediction shown in Table 2. We did the bucket labeling in two stages. CrimeBucket_1 gives zero or non-zero bucketing and CrimeBucket_2 gives different labels for non-zero bucketing.

From the raw crime data we engineered a set of spatio-temporal features described in Table 3 for building the crime volume bucket prediction model.

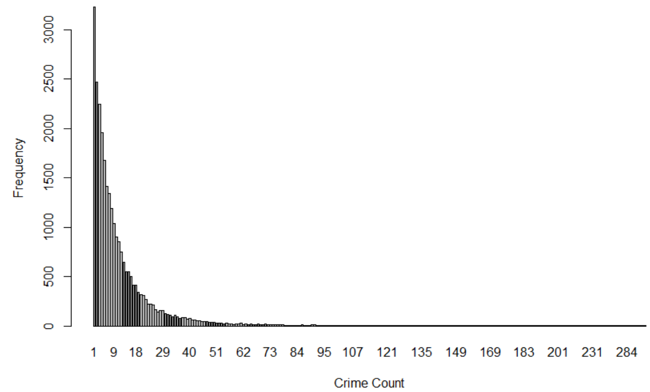

Figure 2: Monthly Crime Volume Distribution

[3]https://www.denvergov.org/opendata/dataset/city-and-county-of-denver-crime

Table 2: Crime Volume Bucketing

Crime Count Range	CrimeBucket_1	CrimeBucket_1
0	zero	
1-7	non-zero	1
8-15	non-zero	2
16-41	non-zero	3
42-103	non-zero	4
>103	non-zero	5

Table 3: Feature Set

Spatio-Temporal Features Engineered for Volume Prediction
1. **Statistical Features**
• In grid mean crime in last 1, 2 and 3 months
• In grid normalized crime in last 1, 2 and 3 months
• Grid rank in last 1, 2 and 3 months
• Grid hotspot frequency in last 1, 2 and 3 months
• Grid hotspot history in last 1, 2 and 3 months
2. **Neighborhood Features**
• Neighbor grids total crime in last 1, 2 and 3 months

6 EXPERIMENTAL RESULTS

We now describe experiments that we have performed for evaluating our proposed approach and compare its performance with the baseline model. In our crime volume prediction first we have built the stage1(zero/non-zero) classifier using the featurized data. We have split the data into 80:20(training and test) based on the months over all years. We have used the following off-the-shelf ML algorithms for training the model: Logistic Regression(LR), Classification and Regression Tree(CART), Random Forest(RF) and XGBoost(XGB). For Stage1 classification the performance of these algorithms are reported in Table 4 . For training the Stage2 classification we first filtered out the training data-points with non-zero CrimeBucket_1 prediction. Then we have used that subset of the training data as the training dataset for stage2 crime volume prediction. For Stage2 volume prediction we have used the same set of 4 ML algorithms. Comparison of the performance metrics for Stage2 classification among these algorithms are reported in Table 6. For testing our volume prediction models first we have run the Stage1 prediction model on test data and have evaluated Stage1 prediction model. Then we have taken the non-zero classified test data-points by Stage1 model and have run Stage2 prediction model on that to get the crime volume predictions for non-zero classes. Among the 4 ML algorithms used in both Stage1 and Stage2 volume prediction XGBoost shows the best performance. In Stage1 prediction XGBoost shows f1-score of 99% on the test dataset. For Stage2 prediction we got an f1-score of 69%. So we took XGBoost as the crime volume prediction model for our approach and take its output to the hotspot prediction algorithm in the next stage of our experiment. For XGBoost

Table 4: Crime Volume Prediction Results in Stage1 (Zero/ Non-Zero Class)

Algorithm	Precision	Recall	F1-Score
LR	0.97	0.95	0.96
CART	0.96	0.96	0.96
RF	0.97	0.97	0.97
XGB	0.99	0.98	0.99

Table 5: Bucket-level Prediction Results in XGB

Bucket	Precision	Recall	F1-Score
Zero	0.94	0.95	0.94
Non-Zero	0.99	0.98	0.99
macro avg	0.99	0.98	0.99

Table 6: Crime Volume Prediction Results in Stage2 (Non-Zero Classes)

Algorithm	Precision	Recall	F1-Score
LR	0.6	0.64	0.62
CART	0.56	0.62	0.59
RF	0.63	0.64	0.64
XGB	0.69	0.69	0.69

Table 7: Bucket-level Prediction Results in XGB

Bucket	Precision	Recall	F1-Score
1	0.83	0.83	0.83
2	0.54	0.57	0.55
3	0.54	0.53	0.53
4	0.45	0.37	0.41
5	0.82	0.79	0.8
macro avg	0.69	0.69	0.69

model the bucket level prediction accuracy numbers for Stage1 are reported in Table 5 and for Stage2 are reported in Table 7. The bucket level prediction accuracy numbers reported for Stage2 tells that for grids with lower(1-7) and higher(>103) crime volume, the performance metrics are higher, where as for grids in the mid buckets the prediction accuracy numbers are considerably less.

In the second stage of our experiment, we have run the crime hotspot prediction using the predicted crime volume label obtained in first stage. We published our numbers for crime hotspot prediction in Table 8. For hotspot prediction our approach shows f1-score of 79%. We compared our numbers with the baseline binary hotspot prediction approach. In baseline binary hotspot prediction we have taken the featurized crime data as input. We have taken the target label as either 1(hotspot) or 0(non-hotspot). We have trained a binary classifier with the same set of spatio-temporal features on the same training dataset and same data partition. Table 8 shows a comparison between the accuracy numbers of the baseline model and our approach. It shows around 5%

Table 8: Crime Hotspot Prediction Results: Baseline Vs. Our Approach

	Precision	Recall	F1-Score
Baseline	0.75	0.73	0.74
Our Approach	0.8	0.77	0.79

Table 9: Class-level Prediction Results in Our Approach

Hotspot Label	Precision	Recall	F1-Score
0	0.98	0.99	0.99
1	0.8	0.77	0.79
macro avg	0.97	0.97	0.97

improvement in precision, 4% improvement in precision and 5% f1-score improvement in our approach. Table 9 gives a detailed class level accuracy numbers for our crime hotspot prediction.

7 CONCLUSION

While crime hotspot prediction is an important ingredient for optimizing the police operation, most studies are limited to static nature of hotspots. In this work, we have considered the problem of predicting the time varying or so called dynamic hotspot using machine learning methodologies. In our approach, we first predict the crime volume on each grid for a monthly time window, where the grids are collectively covering our target area. Then we utilize the grid wise crime volume information to arrive at if a given grid is hotspot or not, for that monthly time window. For the crime volume prediction, rather than predicting absolute crime count, we empirically define a number of crime buckets by using the insights of the grid level crime count distribution. Thus the volume prediction problem is turned out to be predicting the appropriate crime bucket (i.e., multi-class classification problem). We use spatio-temporal features derived from the available crime incident data and address the crime volume prediction through a two stage hierarchical classification - first a binary classification to check if the volume is zero or non-zero. For the grids belonging to non-zero class, we then predict the bucket class (multi- class classification). Noting that the buckets are carrying ordinal information, we use this with the volume prediction result to arrive at if a grid is hotspot or not. We trained and tested the models with real-world crime dataset for Denver, Colorado. In our computational experiment, for crime volume prediction we got f1-score of 69% and for hotspot prediction, f1-score of 79%, on the same dataset. The results show that we got significant improvements in precision, recall and f1-score over the baseline binary hotspot prediction.

As part of future work, we propose to investigate on the prediction for individual category wise crimes. Furthermore, we plan to extend this study to predict for different time window of crime occurrences. These are relatively difficult problems due to the sparse nature of the recorded crime incidents. Moreover, we intend to develop a method of generating an effective police patrolling strategy using the insights from our crime predictive analytics, to enhance the efficiency of the police patrolling system.

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